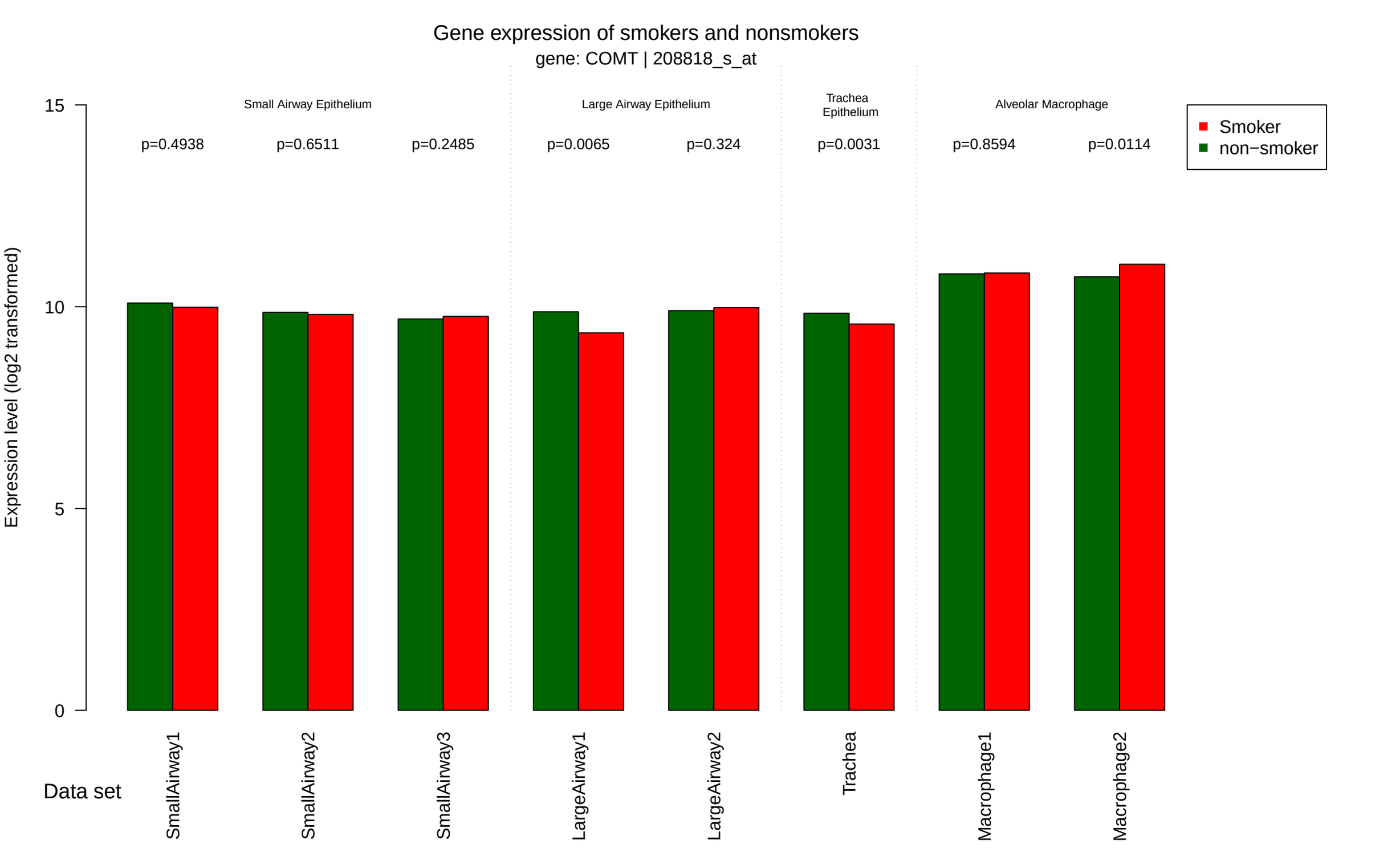
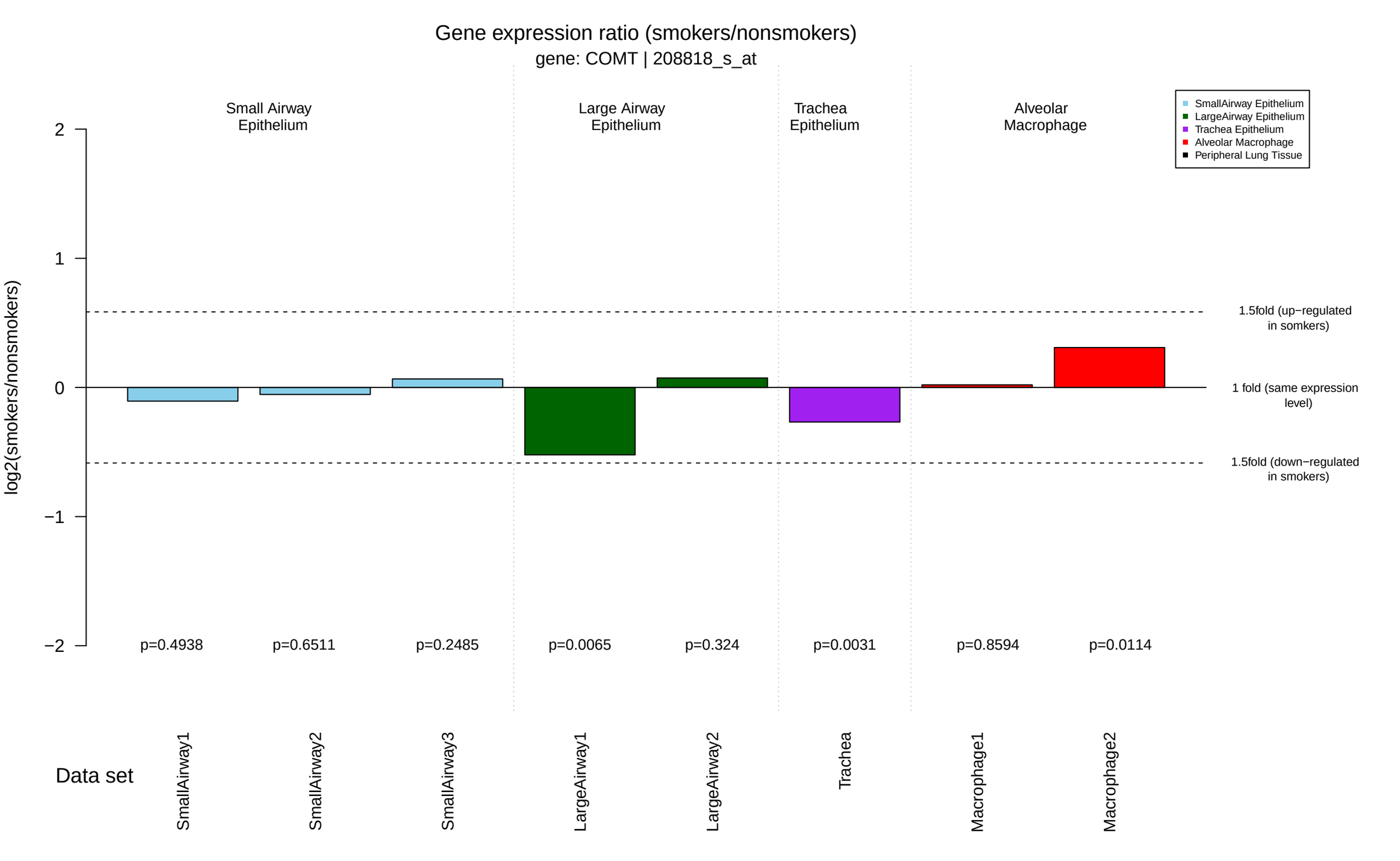
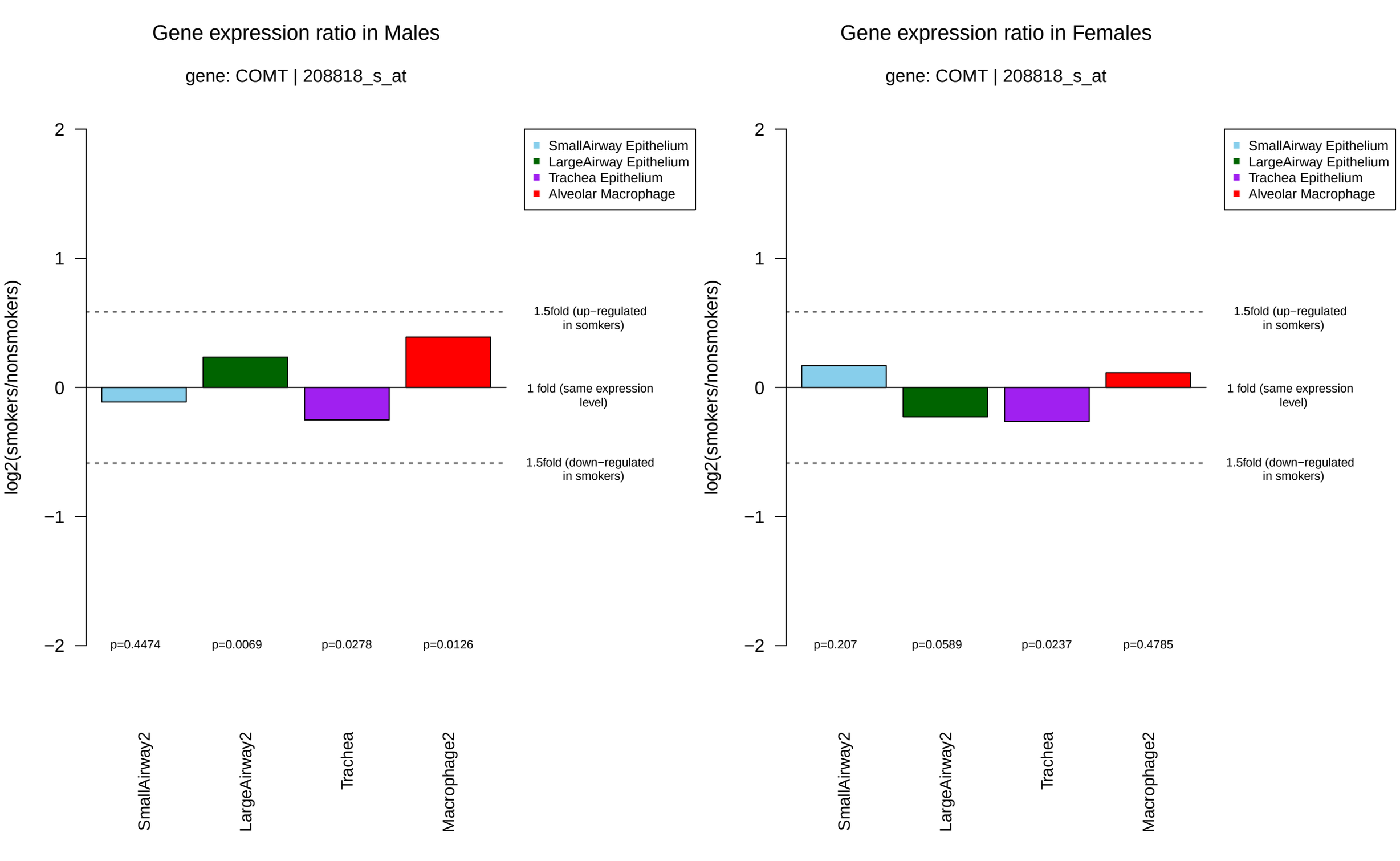
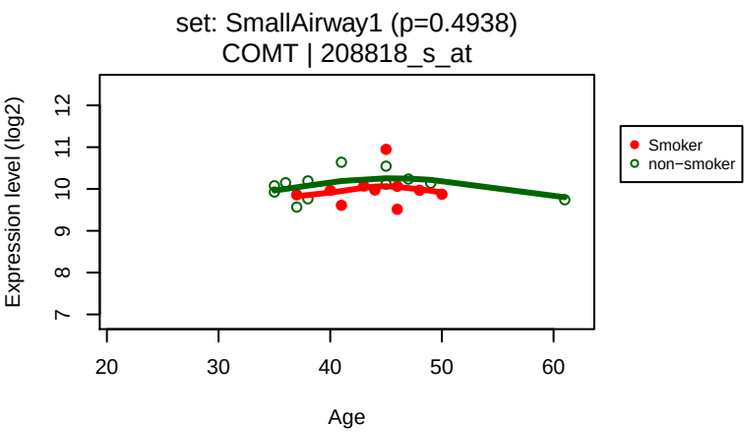
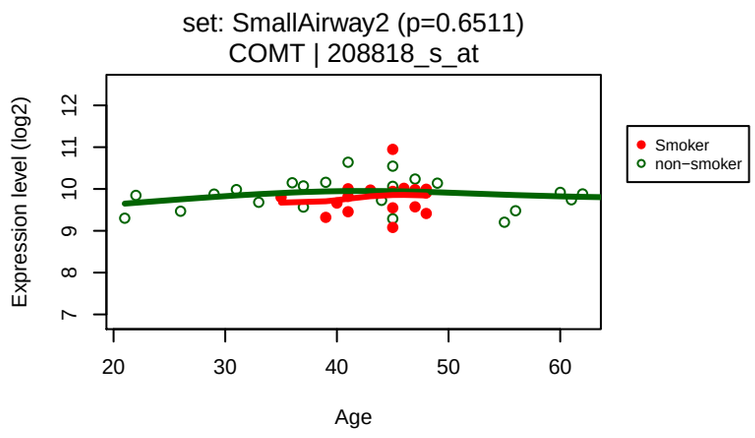
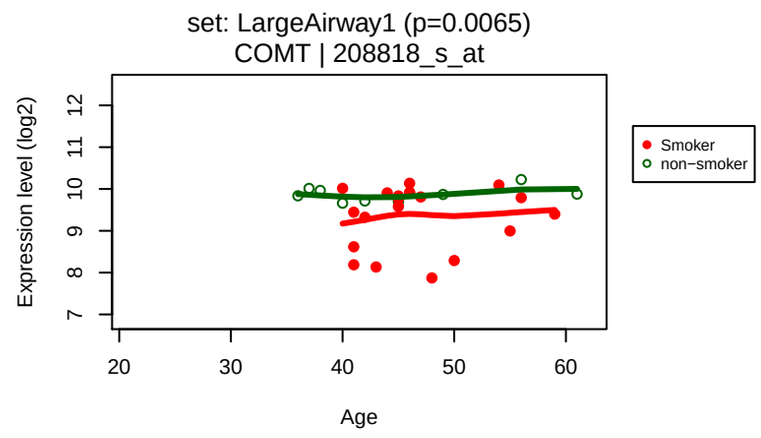

Gene expression of smokers and nonsmokers
gene: COMT | 208818_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.4938
p=0.6511
p=0.2485
p=0.0065
p=0.324
p=0.0031
p=0.8594
p=0.0114
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

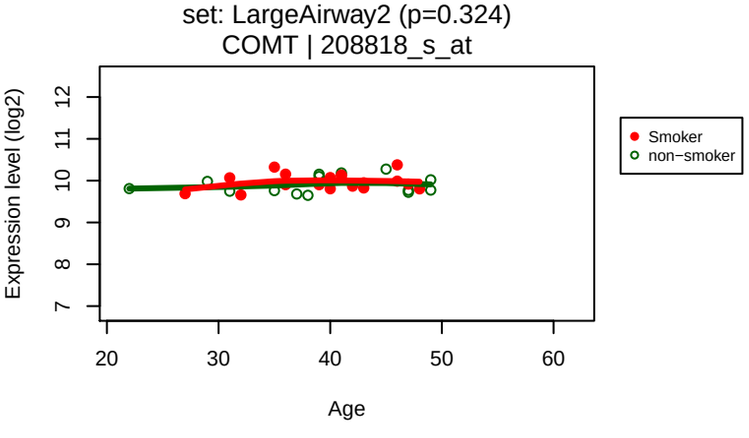

Gene expression ratio (smokers/nonsmokers)
gene: COMT | 208818_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4938
p=0.6511
p=0.2485
p=0.0065
p=0.324
p=0.0031
p=0.8594
p=0.0114
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

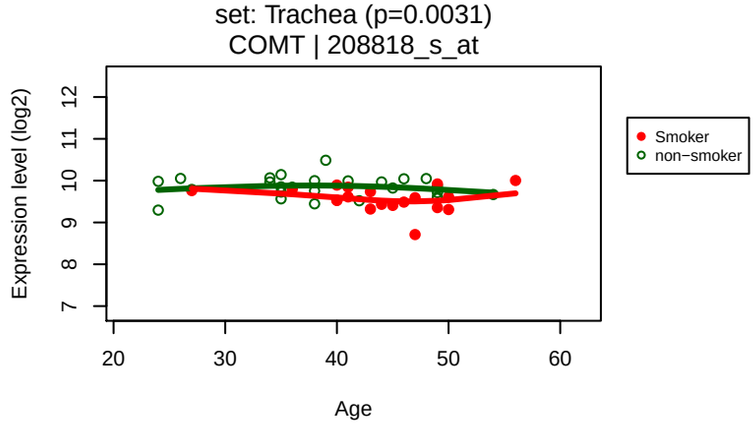

Gene expression ratio in Males
gene: COMT | 208818_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4474
p=0.0069
p=0.0278
p=0.0126
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: COMT | 208818_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.207
p=0.0589
p=0.0237
p=0.4785
SmallAirway2
LargeAirway2
Trachea
Macrophage2

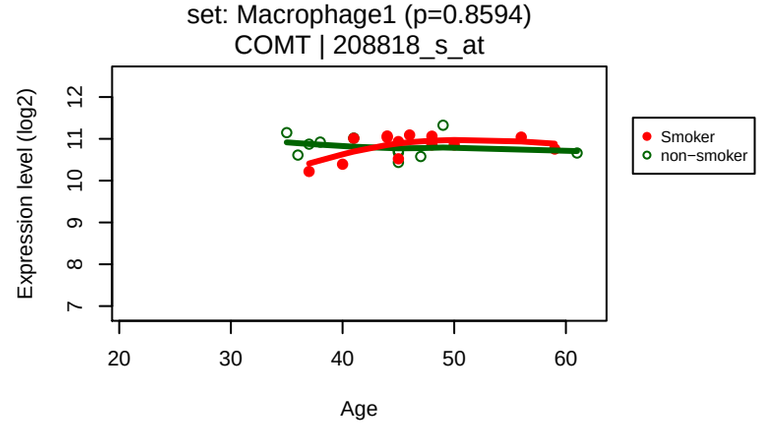

set: SmallAirway1 (p=0.4938)
COMT | 208818_s_at
Expression level (log2)
Age
Smoker
non−smoker

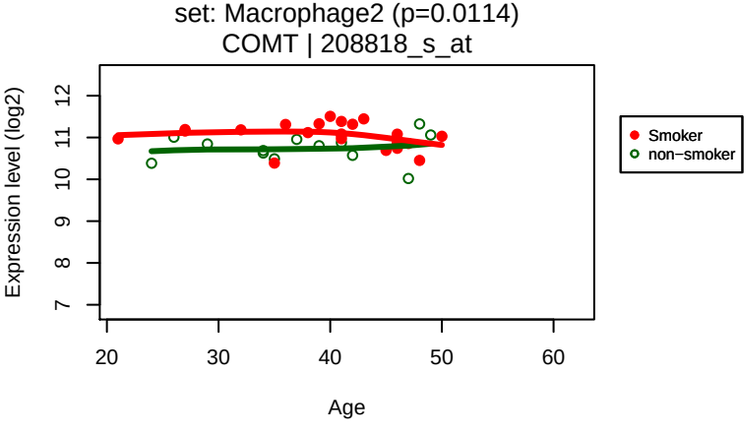

set: SmallAirway2 (p=0.6511)
COMT | 208818_s_at
Expression level (log2)
Age
Smoker
non−smoker

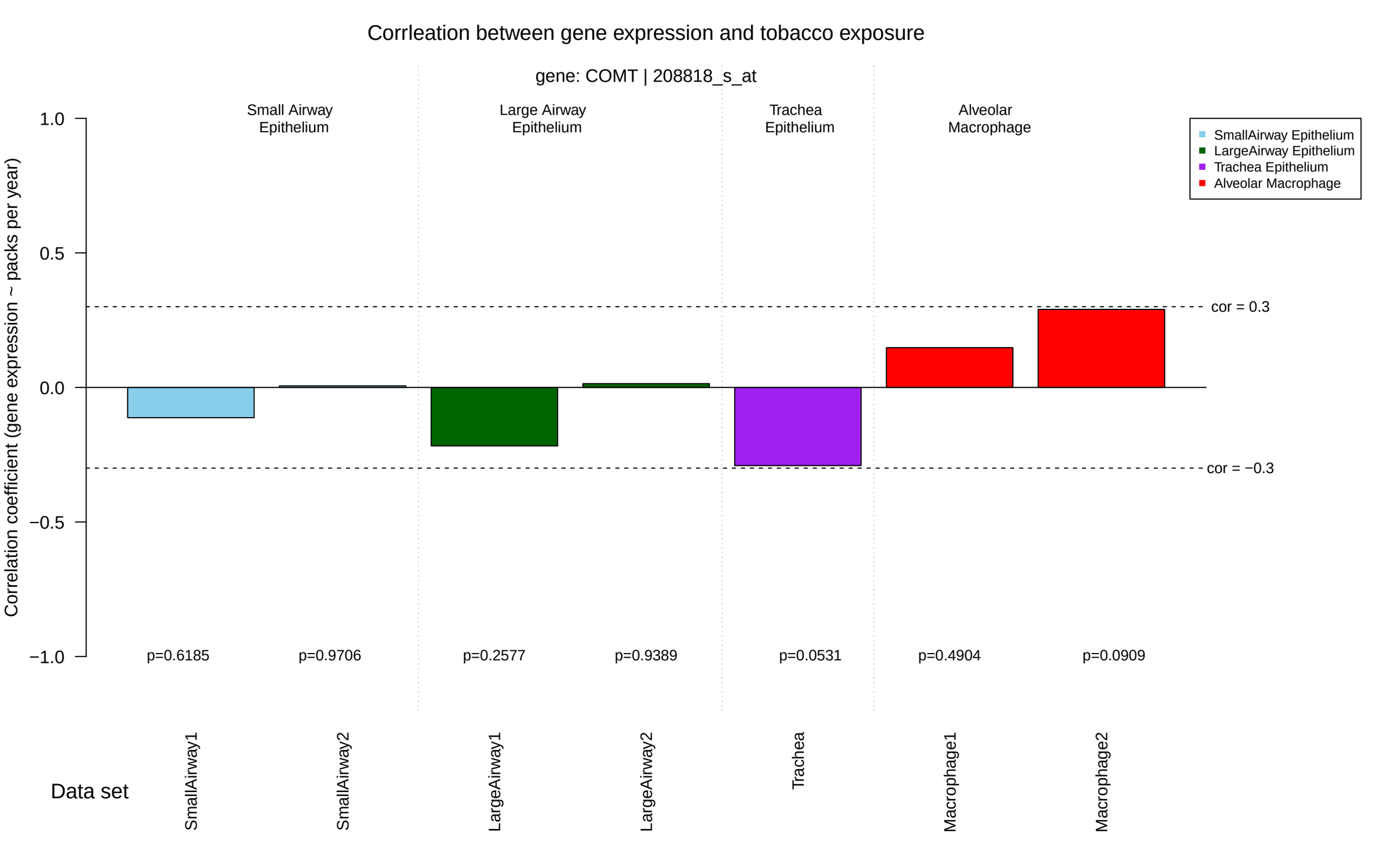

set: LargeAirway1 (p=0.0065)
COMT | 208818_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.324)
COMT | 208818_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0031)
COMT | 208818_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.8594)
COMT | 208818_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0114)
COMT | 208818_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: COMT | 208818_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
Correlation coefficient (gene expression ~ packs per year)
p=0.6185
p=0.9706
p=0.2577
p=0.9389
p=0.0531
p=0.4904
p=0.0909
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2